MTH501 - Linear Algebra (Quiz 04)

Question # 7 of 10 (Start time: 12:44:54 PM, 08 July 2025)

Two vectors u and v in R^n are orthogonal if ------

Select the correct option

◉	$u.v = 0$
○	$u.v \neq 0$

MTH501 - Linear Algebra (Quiz 04)

Question # 3 of 10 (Start time: 12:57:16 PM, 08 July 2025)

If there is a vector $v = (2, 1, 0)$ then $||v||$ is ------

Select the correct option

○	2
○	3
○	0
◉	$\sqrt{5}$

MTH501 - Linear Algebra (Quiz 04)

Question # 5 of 10 (Start time: 12:43:52 PM, 08 July 2025)

If two rows are orthogonal, they are _____.

Select the correct option

- (●) linearly independent
- ( ) linearly dependent

MTH501 - Linear Algebra (Quiz 04)

Question # 5 of 10 (Start time: 12:58:07 PM, 08 July 2025)

The norm of v is the non-negative scalar $||v||$ defined by

Select the correct option

- $||v|| = \sqrt{v_1^2 + v_2^2 + \ldots v_n^2}$
- $||v|| = v_1^2 + v_2^2 + \ldots v_n^2$

MTH501 - Linear Algebra (Quiz 04)

Question # 6 of 10 (Start time: 12:44:16 PM, 08 July 2025)

If x is _____ to both u and v, then x must be orthogonal to u – v.

Select the correct option

◉	orthogonal
○	orthonormal

MTH501 - Linear Algebra (Quiz 04)

Question # 9 of 10 (Start time: 12:59:20 PM, 08 July 2025)

If x is _ _ _ _ _ _ _ to both u and v, then x must be orthogonal to u − v.

Select the correct option

◉	orthogonal
○	orthonormal

MTH501 - Linear Algebra (Quiz 04)

Question # 6 of 10 (Start time: 12:58:36 PM, 08 July 2025)

Any finite dimensional inner product space possesses an orthonormal basis

Select the correct option

◉	True
○	False

MTH501 - Linear Algebra (Quiz 04)

Question # 3 of 10 (Start time: 12:52:52 PM, 08 July 2025)

Suppose x,y,z are some vectors in an inner product space (v,<,>) such that <x,y>=<x,z> for all x∈V, then y=z

Select the correct option

○	False
●	True

MTH501 - Linear Algebra (Quiz 04)

Question # 10 of 10 (Start time: 12:46:12 PM, 08 July 2025)

A square matrix with _____ columns is an orthogonal matrix.

Select the correct option

○	orthogonal
○	same
●	orthonormal
○	even

MTH501 - Linear Algebra (Quiz 04)

Question # 10 of 10 (Start time: 12:55:00 PM, 08 July 2025)

If the columns of a matrix are _____ then the matrix is invertible.

Select the correct option

○	linearly dependent
◉	linearly independent

MTH501 - Linear Algebra (Quiz 04)

Question # 1 of 10 (Start time: 12:56:13 PM, 08 July 2025)

If A is m\cross n and b is in R^m, a least square solution of $Ax = b$ is an $\widehat{x}$ in R^n such that

Select the correct option

○	$\|\|b - A\widehat{x}\|\| >= \|\|b - Ax\|\| \forall x \in R^n$
◉	$\|\|b - A\widehat{x}\|\| <= \|\|b - Ax\|\| \forall x \in R^n$

MTH501 - Linear Algebra (Quiz 04)

Question # 8 of 10 (Start time: 12:54:28 PM, 08 July 2025)

An n x n matrix A is invertible if and only if _____ is not an eigen value of A.

Select the correct option

○	1
◉	0
○	2
○	-1

Question # 5 of 10 (Start time: 12:53:28 PM, 08 July 2025)

Let W be a subspace of R^n, then each y in R^n can be written uniquely in the form

Select the correct option

◉	$y = \hat{y} + z$
○	none of the above
○	$y = \hat{y}/z$
○	$y = \hat{y} - z$

MTH501 - Linear Algebra (Quiz 04)

Question # 7 of 10 (Start time: 12:54:10 PM, 08 July 2025)

$7x + 2y = 16$
$-21x - 6y = 24$
The above system has a _____ solution.

Select the correct option

○	unique
○	trivial
○	consistent
◉	inconsistent

MTH501 - Linear Algebra (Quiz 04)

Question # 1 of 10 (Start time: 12:42:18 PM, 08 July 2025)

If we divide a non-zero vector by its length we get a

Select the correct option

○	normalized vector
○	zero vector
○	none of these
◉	unit vector

MTH501 - Linear Algebra (Quiz 04)

Question # 2 of 10 (Start time: 12:42:42 PM, 08 July 2025)

The two vectors are said to be equivalent if:

Select the correct option

◉	Both (a) and (b).
○	Same direction
○	None of the above
○	Same length.

MTH501 - Linear Algebra (Quiz 04)

Question # 8 of 10 (Start time: 12:59:06 PM, 08 July 2025)

The product of upper triangular matrices is

Select the correct option

○	Diagonal matrix.
○	None of the above
○	Lower triangular matrix.
◉	Upper triangular matrix

MTH501 - Linear Algebra (Quiz 04)

Question # 3 of 10 (Start time: 12:43:01 PM, 08 July 2025)

Elementary row operations on a matrix do not affect the -------- relations among the columns of the matrix.

Select the correct option

- (●) Linear dependence
- ( ) Linear independence

MTH501 - Linear Algebra (Quiz 04)

Question # 4 of 10 (Start time: 12:57:47 PM, 08 July 2025)

If u and v are non zero vectors in either R^2 or R^3 then by the law of cosines $||u-vII^2$=---------

Select the correct option

- (●) $||u-v||^2 = ||u||^2 + ||v||^2 - 2||u||||v||cos\theta$
- ( ) $||u-v||^2 = ||u||^2 + ||v||^2 + 2||u||||v||cos\theta$

MTH501 - Linear Algebra (Quiz 04)

Question # 7 of 10 (Start time: 12:58:50 PM, 08 July 2025)

Elementary row operations on a matrix do not affect the -------- relations among the columns of the matrix.

Select the correct option

- (●) Linear dependence
- ( ) Linear independence

MTH501 - Linear Algebra (Quiz 04)

Question # 4 of 10 (Start time: 12:43:25 PM, 08 July 2025)

Let W is a finite dimensional subspace of an inner product space V and y is any vector in V. The best approximation to y from W is then

Select the correct option

- ( ) $Proj_{w}^{y}$
- (●) $Proj_{y}^{w}$

MTH501 - Linear Algebra (Quiz 04)

Question # 1 of 10 (Start time: 12:51:56 PM, 08 July 2025)

Let

$$s = \{u_1 + u_2 + \ldots\ldots u_p\}$$

be a basis for a subspace W of

$$R^n$$

, is also an orthogonal basis if S is an orthogonal set.

Select the correct option

○	True
◉	False

MTH501 - Linear Algebra (Quiz 04)

Question # 10 of 10 (Start time: 12:59:36 PM, 08 July 2025)

Let

$$s = \{u_1 + u_2 + \ldots\ldots u_p\}$$

be a basis for a subspace W of

$$R^n$$

, is also an orthogonal basis if S is an orthogonal set.

Select the correct option

○	True
◉	False

MTH501 - Linear Algebra (Quiz 04)

Question # 4 of 10 (Start time: 12:53:08 PM, 08 July 2025)

If a _____ matrix has orthonormal columns, then it also has orthonormal rows.

Select the correct option

○	singular
◉	square
○	diagonal
○	rectangular

MTH501 - Linear Algebra (Quiz 04)

Question # 9 of 10 (Start time: 12:45:46 PM, 08 July 2025)

Let W be a subspace of R^n, then each y in R^n can be written uniquely in the form

Select the correct option

◉	$y = \hat{y} + z$
○	$y = \hat{y} - z$
○	none of the above
○	$y = \hat{y}/z$

Question # 2 of 10 (Start time: 12:56:36 PM, 08 July 2025)

If $\{u_1, u_2, \ldots, u_p\}$ is an orthonormal basis for a subspace W of R^n, then ...

Select the correct option

○	$Proj_w y = (y.u_1)u_1 + (y.u_2)u_2 + \ldots + (y.u_p)u_p$
◉	$Proj_w y = (y.u_1)u_p + (y.u_2)u_p + \ldots + (y.u_p)u_p$

Question # 6 of 10 (Start time: 12:53:46 PM, 08 July 2025)

If

$$s = \{u_1 + u_2 + \ldots\ldots u_p\}$$

is an orthogonal basis for a subspace W of

$$R^n$$

Then each y in W can be uniquely expressed as a linear combination of

$$u_1 + u_2 + \ldots\ldots u_p$$

That is

$$c_1 u_1 + c_2 u_2 + \ldots\ldots c_p u_p$$

Where

$$\frac{y.\,u_j}{u_j.\,u_j}$$

Select the correct option

○	True
◉	False

MTH501 - Linear Algebra (Quiz 04)

Question # 8 of 10 (Start time: 12:45:16 PM, 08 July 2025)

If B=

$$\{v_1, v_2, v_3\}$$

is an orthogonal set of vectors with respect to an inner product on a vector space V, then the set B

Select the correct option

○	Linearly dependent.
○	Spans the vector space v
○	Is an orthonormal basis for v
◉	Is linearly independent.

MTH501 - Linear Algebra (Quiz 04) Quiz Start Ti

Question # 2 of 10 (Start time: 12:52:24 PM, 08 July 2025)

Let W be a subspace of R^n, y be any vector in R^n and $\hat{y}$ the orthogonal projection of y onto W. Then $\hat{y}$ is the closest point in W to y, in the sense that

Select the correct option

◉	$\|\|y - \hat{y}\|\| < \|\|y - v\|\|$
○	$\|\|y - \hat{y}\|\| > \|\|y - v\|\|$

Click to Save Answer

MTH501 - Linear Algebra (Quiz 04)

Question # 9 of 10 (Start time: 12:54:45 PM, 08 July 2025)

If the augmented matrices of two linear systems are row equivalent, then the two systems have the _____ solution set.

Select the correct option

- ○ different
- ◉ same
- ○ inconsistant
- ○ trivial

Cl